Farrukh Iqbal Khan

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Education

MS Computer Engineering	NUST, EME College, Rawalpindi	2011-2014
BE Electronics Engineering	NUST, SEECS, Islamabad	2006-2010
HSSC (Pre-Engineering)	Superior College, Lahore	2004-2006
SSC	Divisional Public School, Lahore	2004

Technical Skills

Languages:

- C++
- C
- Matlab
- Verilog

Libraries:

- OpenCV

Software Tools:

- Git
- Microsoft Visio 2013
- Microsoft Visual Studio 2012
- Xilinx ISE
- Asterisk PBX
- Matlab

Operating systems:

- Linux (Ubuntu)
- Windows(XP, 7, 8)

Academic Projects

- **MS Thesis**

"Multi modal registration of thermo-visual images"

Abstract: Image registration is a method in computer vision which maps corresponding features from multiple images so that they can be aligned together using homography estimation. In the thesis, an improvement on SURF method of feature detection and description for registration of Thermo-Visual images is proposed. Currently, I am reviewing the literature and conducting related experiments.

- **Post-Graduate Projects**
  - **"Image up scaling using Iterative Curvature Based Interpolation**", drafted a research paper for *Digital Image Processing* course.
  - **"Super Resolution using Linear Regression"** drafted a research paper using Generalized Linear Model for *Computer Vision* course.
  - **"Design & implementation of a pipelined Math processor using Verilog HDL"**, implemented and verified the design on Spartan-3 FPGA for *Computer Architecture* course.
  - "**A comparison of K-Nearest Neighbor Algorithm & Naïve Bayes Classifier"**, research project carried out for *Pattern Recognition* course.
  - **"Speech synthesis using Linear Predictive coding (LPC)"**, implemented the design using Prony's algorithm on Matlab.

- **Undergraduate Projects**
  - Final Year Project: **"Implementation of an IP based communication system for providing value added services using Asterisk PBX"**. The provided services included interactive voice response (IVR), call conferencing, voicemail, presence call routing.
  - Semester Projects:
    - Police Vehicle Command and Control System
    - Library Management System for College
    - Design of nMOS Field Effect Transistor using Cadence Tools

Research Interests

I have interest in following research topics.

- Multimodal image registration
- Local image feature detection and description
- Super resolution, image segmentation
- Multi resolution analysis
- Harmonic analysis (Fourier & wavelet domain processing)